

Calc 1 Lecture 19: L'hospital's rule

Now that we have more tools at our disposal, we revisit limits.

Consider the following limits:

- | | | | |
|--|--|---------------------------------|------------------------------------|
| a) $\lim_{x \rightarrow 1^+} \frac{x}{\ln(x)} = \infty$ | numerator = $x \rightarrow 1$
denominator = $\ln(x) \rightarrow 0^+$ | " $\frac{1}{0^+} = \infty$ " | } determinate |
| b) $\lim_{x \rightarrow 0^+} \frac{x+1}{\ln(x)} = 0$ | numerator = $x+1 \rightarrow 1$
denominator = $\ln x \rightarrow -\infty$ | " $\frac{1}{-\infty} = 0$ " | |
| c) $\lim_{x \rightarrow 0^+} \frac{x}{\ln(x)} = 0$ | numerator = $x \rightarrow 0$
denominator = $\ln x \rightarrow -\infty$ | " $\frac{0}{-\infty} = 0$ " | |
| d) $\lim_{x \rightarrow \infty} \frac{\ln(x)}{1 + \frac{1}{x}} = \infty$ | numerator = $\ln x \rightarrow \infty$
denom. = $1 + \frac{1}{x} \rightarrow 1$ | " $\frac{\infty}{1} = \infty$ " | |
| e) $\lim_{x \rightarrow \infty} \frac{\ln(x)}{x^{-1/2}} = \infty$ | numerator = $\ln x \rightarrow \infty$
denom. = $x^{-1/2} = \frac{1}{\sqrt{x}} \rightarrow 0$ | " $\frac{\infty}{0} = \infty$ " | } indeterminate
no general rule |
| f) $\lim_{x \rightarrow 1^+} \frac{\ln(x)}{x-1} = ?$ | numerator = $\ln x \rightarrow 0$
denom. = $x-1 \rightarrow 0$ | " $\frac{0}{0} = ?$ " | |
| g) $\lim_{x \rightarrow \infty} \frac{\ln(x)}{x} = ?$ | numerator = $\ln x \rightarrow \infty$
denom. = $x \rightarrow \infty$ | " $\frac{\infty}{\infty} = ?$ " | |

observations:

When evaluating the limit of a quotient of functions, we can easily evaluate limits of the form

" $\frac{\text{bounded}}{\infty} = 0$ " " $\frac{\infty}{\text{bounded}} = \infty$ " " $\frac{0}{\text{bounded}} = 0$ " etc.

but we cannot evaluate " $\frac{0}{0}$ " or " $\frac{\infty}{\infty}$ " without further info.

Note: We've seen indeterminate forms before, e.g.

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} \quad \left. \begin{array}{l} \text{numerator} \rightarrow 0 \\ \text{denom.} \rightarrow 0 \end{array} \right\} \text{"0/0" indeterminate} = \lim_{x \rightarrow 1} \frac{\cancel{(x-1)}(x+1)}{\cancel{x-1}} = \boxed{2}$$

or

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} \quad \left. \begin{array}{l} \text{numerator} \rightarrow 0 \\ \text{denom} \rightarrow 0 \end{array} \right\} \text{"0/0" indeterminate} = \boxed{1}$$

But these ideas can't always be used.

L'Hospital's rule gives a general method for evaluating limits of such indeterminate forms.

Definition: a limit $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ is of indeterminate form if:

either "0/0": $\lim_{x \rightarrow a} f(x) = 0, \quad \lim_{x \rightarrow a} g(x) = 0$

or " $\frac{\infty}{\infty}$ ": $\lim_{x \rightarrow a} f(x) = \infty, \quad \lim_{x \rightarrow a} g(x) = \infty$

L'Hospital's rule suppose f and g are differentiable

and $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ is of indeterminate form.

$$\text{Then } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

if the limit on the right exists.

Note: in the above, " $x \rightarrow a$ " can be replaced with " $x \rightarrow \infty$ " or " $x \rightarrow a^+$ " or " $x \rightarrow a^-$ "

Example Find $\lim_{x \rightarrow 1} \frac{\ln(x)}{x-1}$

Step 1: check if indeterminate

numerator: $\lim_{x \rightarrow 1} \ln(x) = \ln(1) = 0 \Rightarrow \frac{0}{0}$ indeterminate

denominator: $\lim_{x \rightarrow 1} (x-1) = 1-1 = 0$

Step 2: Apply L'Hospital's

$\lim_{x \rightarrow 1} \frac{\ln(x)}{x-1} = \lim_{x \rightarrow 1} \frac{\frac{d}{dx}(\ln x)}{\frac{d}{dx}(x-1)} = \lim_{x \rightarrow 1} \frac{1/x}{1} = \lim_{x \rightarrow 1} \frac{1}{x} = 1$

↑ L'Hospital's

direct subst. $x=1$

Answer: $\lim_{x \rightarrow 1} \frac{\ln(x)}{x-1} = \boxed{1}$

Explanation of the " $\frac{0}{0}$ " case of L'Hospital's rule:

think about linear approximations

$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} \frac{0}{0} \Rightarrow \underbrace{f(a)=0}, \underbrace{g(a)=0}$

as x is close to a , $f(x) \approx \cancel{f(a)} + f'(a)(x-a) = f'(a)(x-a)$
 $g(x) \approx \cancel{g(a)} + g'(a)(x-a) = g'(a)(x-a)$

$\frac{f(x)}{g(x)} \approx \frac{f'(a)(x/a)}{g'(a)(x/a)} = \frac{f'(a)}{g'(a)}$

Onto the " $\frac{0}{0}$ " case...

Example Calculate $\lim_{x \rightarrow \infty} \frac{e^x}{x^2}$

Step 1: check indeterminate.

numerator: $\lim_{x \rightarrow \infty} e^x = \infty$

denominator: $\lim_{x \rightarrow \infty} x^2 = \infty$

" $\frac{\infty}{\infty}$ " indeterminate

Step 2: use L'Hospital's.

$$\lim_{x \rightarrow \infty} \frac{e^x}{x^2} = \lim_{x \rightarrow \infty} \frac{d/dx(e^x)}{d/dx(x^2)} = \lim_{x \rightarrow \infty} \frac{e^x}{2x}$$

$\left. \begin{array}{l} e^x \rightarrow \infty \\ 2x \rightarrow \infty \end{array} \right\}$ still indeterminate

use L'Hospital's again:

$$\lim_{x \rightarrow \infty} \frac{e^x}{2x} = \lim_{x \rightarrow \infty} \frac{\frac{d}{dx}(e^x)}{\frac{d}{dx}(2x)} = \lim_{x \rightarrow \infty} \frac{e^x}{2} = \boxed{\infty}$$

Note: more generally, $\lim_{x \rightarrow \infty} \frac{e^x}{\text{Polynomial}} = \infty$

$$\lim_{x \rightarrow \infty} \frac{\text{Polynomial}}{e^x} = 0$$

" e^x grows much faster in the long run than any polynomial"

observation:

Sometimes multiple applications of L'Hospital's rule are necessary before we simplify to a non-indeterminate form.

Example

Calculate $\lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt[3]{x}}$

Step 1: check if indeterminate

numerator: $\lim_{x \rightarrow \infty} \ln x = \infty$
denom: $\lim_{x \rightarrow \infty} \sqrt[3]{x} = \infty$ } " $\frac{\infty}{\infty}$ " indeterminate form

Step 2: L'Hospital's:

$$\lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt[3]{x}} = \lim_{x \rightarrow \infty} \frac{\frac{d}{dx}(\ln x)}{\frac{d}{dx}(x^{1/3})} = \lim_{x \rightarrow \infty} \frac{1/x}{\frac{1}{3}x^{-2/3}}$$

$$= \lim_{x \rightarrow \infty} \frac{1/x}{1/3 x^{2/3}} = \lim_{x \rightarrow \infty} \frac{1}{x} \cdot 3x^{2/3} = \lim_{x \rightarrow \infty} \frac{3}{x^{1/3}} = \boxed{0}$$

observation

$\ln x$ grows much more slowly than x^a for any positive a .

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x^a} = 0$$

Example Find $\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3}$

Step 1: check indeterminate.

$$\begin{array}{lcl} \text{as } x \rightarrow 0: & \tan x - x \rightarrow \tan(0) - 0 = 0 & \\ & x^3 \rightarrow 0^3 = 0 & \end{array} \left. \vphantom{\begin{array}{lcl} \text{as } x \rightarrow 0: & \tan x - x \rightarrow \tan(0) - 0 = 0 & \\ & x^3 \rightarrow 0^3 = 0 & \end{array}} \right\} \begin{array}{l} \text{"0/0"} \\ \text{indeterminate} \end{array}$$

Step 2: L'hospital's.

$$\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3} = \lim_{x \rightarrow 0} \frac{\frac{d}{dx}(\tan x - x)}{\frac{d}{dx}(x^3)} = \lim_{x \rightarrow 0} \frac{\sec^2 x - 1}{3x^2}$$

still indeterminate

Approach 1: do L'hospital's again

Approach 2: trig identity $\sec^2 x - 1 = \tan^2 x$

$$\lim_{x \rightarrow 0} \frac{\sec^2 x - 1}{3x^2} = \lim_{x \rightarrow 0} \frac{\tan^2 x}{3x^2} = \frac{1}{3} \lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^2$$

$$= \frac{1}{3} \left(\lim_{x \rightarrow 0} \frac{\tan x}{x} \right)^2 = \frac{1}{3} \cdot 1^2 = \boxed{\frac{1}{3}}$$

L'hospital's or $\lim_{x \rightarrow 0} \frac{\tan x}{x} = \lim_{x \rightarrow 0} \frac{\sin x}{\cos x \cdot x}$

$$= \boxed{1}$$

if we did approach 1:

$$\lim_{x \rightarrow 0} \frac{\sec^2 x - 1}{3x^2} = \lim_{x \rightarrow 0} \frac{2 \sec x \cdot \sec x \tan x}{6x}$$

L'Hospital's

$$= \lim_{x \rightarrow 0} \frac{\sec^2 x \tan x}{3x} \quad \left. \begin{array}{l} \} \rightarrow 0 \\ \} \rightarrow 0 \end{array} \right\}$$

still indeterminate

$$= \underbrace{\left(\lim_{x \rightarrow 0} \frac{\sec^2 x}{3} \right)}_{\frac{1}{3} \text{ by direct subst.}} \cdot \underbrace{\left(\lim_{x \rightarrow 0} \frac{\tan x}{x} \right)}_{\substack{\text{L'Hospital's} \\ \text{or trig limit}}} = \frac{1}{3} \cdot 1 = \boxed{\frac{1}{3}}$$

Example Find $\lim_{x \rightarrow \pi^-} \frac{\sin x}{1 - \cos x}$

Wrong approach:

~~$$\lim_{x \rightarrow \pi^-} \frac{\cos x}{\sin x} = \frac{-1}{0^+} = -\infty$$~~

check indeterminate:

$$\sin(\pi) = 0$$

$$1 - \cos(\pi) = 1 - (-1) = 2$$

$$\frac{0}{2} = \boxed{0}$$

Other indeterminate forms

• Indeterminate products: " $0 \cdot \infty$ "

general strategy: make it look like a quotient. $f \cdot g = \frac{f}{1/g}$
then use L'Hospital's

Example: Evaluate $\lim_{x \rightarrow 0^+} \underbrace{x}_{\rightarrow 0} \underbrace{\ln x}_{\rightarrow -\infty}$

$$\lim_{x \rightarrow 0^+} x = 0, \quad \lim_{x \rightarrow 0^+} \ln x = -\infty \quad \Rightarrow \text{"}0 \cdot (-\infty)\text{" indeterminate}$$

rewrite $x \cdot \ln x$ as a quotient: $\frac{x}{1/\ln x}$ or $\frac{\ln x}{1/x}$
easier

$$\lim_{x \rightarrow 0^+} x \ln x = \lim_{x \rightarrow 0^+} \frac{\ln x}{1/x} \quad \left. \begin{array}{l} \} \rightarrow -\infty \\ \} \rightarrow \infty \end{array} \right\} \text{indeterminate "}\frac{\infty}{\infty}\text{"}$$

$$\begin{aligned} &\text{L'Hospital's} \\ &= \lim_{x \rightarrow 0^+} \frac{\frac{d}{dx}(\ln x)}{\frac{d}{dx}\left(\frac{1}{x}\right)} = \lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} = \lim_{x \rightarrow 0^+} \left(\frac{1}{x} \cdot -x^2\right) \end{aligned}$$

$$= \lim_{x \rightarrow 0^+} -x = \boxed{0}$$

• Indeterminate differences: " $\infty - \infty$ " " $(\infty + (-\infty))$ "

general strategy: try to rewrite as a quotient, use L'Hospital's

Example Compute $\lim_{x \rightarrow \frac{\pi}{2}^-} (\sec x - \tan x)$

Step 1: check indeterminate

$$\lim_{x \rightarrow \frac{\pi}{2}^-} \sec x = \lim_{x \rightarrow \frac{\pi}{2}^-} \frac{1}{\cos x} = \frac{1}{0^+} = \infty$$

$$\lim_{x \rightarrow \frac{\pi}{2}^-} \tan x = \lim_{x \rightarrow \frac{\pi}{2}^-} \frac{\sin x}{\cos x} = \frac{1}{0^+} = \infty$$

$$\lim_{x \rightarrow \frac{\pi}{2}^-} (\sec x - \tan x) = \infty - \infty \text{ indeterminate}$$

Step 2: convert to a quotient

$$\lim_{x \rightarrow \frac{\pi}{2}^-} (\sec x - \tan x) = \lim_{x \rightarrow \frac{\pi}{2}^-} \left(\frac{1}{\cos x} - \frac{\sin x}{\cos x} \right)$$

$$= \lim_{x \rightarrow \frac{\pi}{2}^-} \left(\frac{1 - \sin x}{\cos x} \right) \begin{matrix} \text{ } \end{matrix} \left. \begin{matrix} \text{ } \end{matrix} \right\} \begin{matrix} \rightarrow 0 \\ \rightarrow 0 \end{matrix} \text{ "0/0"}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}^-} \frac{\frac{d}{dx}(1 - \sin x)}{\frac{d}{dx}(\cos x)} = \lim_{x \rightarrow \frac{\pi}{2}^-} \frac{-\cos x}{-\sin x} = \frac{0}{-1} = \boxed{0}$$

direct substitution

Example Find $\lim_{x \rightarrow \infty} \left(\underbrace{x^2}_{\downarrow \infty} - \underbrace{x \ln(x)}_{\downarrow \infty} \right)$ "∞ - ∞"

Try to multiply and divide by x^2 (equivalently, factor out an x^2)

$$x^2 - x \ln(x) = \frac{x^2 \cdot (x^2 - x \ln x)}{x^2} = x^2 \left(\underbrace{\frac{x^2}{x^2}}_1 - \frac{x \ln x}{x^2} \right)$$

$$= \underbrace{x^2}_{\downarrow \infty} \cdot \underbrace{\left(1 - \frac{\ln x}{x} \right)}_{\downarrow 1-0=1}$$

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x} = ? \text{ "}\frac{\infty}{\infty}\text{"}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{d}{dx}(\ln x)}{\frac{d}{dx}(x)} = \lim_{x \rightarrow \infty} \frac{1/x}{1} = 0$$

Answer: $\lim_{x \rightarrow \infty} x^2 - x \ln(x) = \lim_{x \rightarrow \infty} \underbrace{x^2}_{\downarrow \infty} \cdot \underbrace{\left(1 - \frac{\ln x}{x} \right)}_{\downarrow 1}$

$$= \text{"}\infty \cdot 1\text{"} = \boxed{\infty}$$

- indeterminate powers $1^\infty, \infty^0, 0^0$

general strategy: take $\ln$, convert to product, go from there

Example: Calculate $\lim_{x \rightarrow 0^+} (1 + \sin(4x))^{\cot x}$

Step 1: check indeterminate.

$$\text{as } x \rightarrow 0^+, \quad \left. \begin{array}{l} (1 + \sin(4x)) \rightarrow 1 + \sin(0) = 1 \\ \cot x = \frac{\cos x}{\sin x} \rightarrow \frac{1}{0^+} \rightarrow \infty \end{array} \right\} \text{"} \infty^0 \text{" indeterminate}$$

Step 2: Apply $\ln$.

$$\begin{aligned} \ln L &= \lim_{x \rightarrow 0^+} (1 + \sin(4x))^{\cot x} \\ \ln L &= \ln \left(\lim_{x \rightarrow 0^+} (1 + \sin(4x))^{\cot x} \right) = \lim_{x \rightarrow 0^+} \left(\ln \left((1 + \sin(4x))^{\cot x} \right) \right) \\ \text{log Property} \quad &= \lim_{x \rightarrow 0^+} [\cot x \cdot \ln(1 + \sin(4x))] \\ &= \lim_{x \rightarrow 0^+} \left(\frac{\ln(1 + \sin(4x))}{1/\cot x} \right) = \lim_{x \rightarrow 0^+} \left(\frac{\ln(1 + \sin(4x))}{\tan x} \right) \quad \begin{array}{l} \ln(1) = 0 \\ \tan 0 = 0 \end{array} \\ \text{L'Hospital's} \quad &= \lim_{x \rightarrow 0^+} \left(\frac{\frac{d}{dx} (\ln(1 + \sin(4x)))}{\frac{d}{dx} \tan x} \right) = \lim_{x \rightarrow 0} \frac{\frac{d}{dx} (1 + \sin(4x)) \cdot 4 \cos(4x)}{1 + \sin(4x)} \cdot \frac{1}{\sec^2 x} \end{aligned}$$

$\frac{d}{dx} (\ln f) = \frac{f'}{f}$

f

$$= \lim_{x \rightarrow 0^+} \frac{4 \cos(4x)}{(1 + \sin(4x)) \sec^2 x} = \frac{4 \cdot 1}{(1+0) \cdot 1^2} = \boxed{4}$$

↑
direct
substitution

$$\ln L = 4 \Rightarrow \boxed{L = e^4}$$

Example Find $\lim_{x \rightarrow 0^+} x^x$

" 0^0 " indeterminate

$$L = \lim_{x \rightarrow 0^+} x^x$$

ln



$$\ln L = \lim_{x \rightarrow 0^+} (\ln(x^x)) = \lim_{x \rightarrow 0^+} (x \ln x)$$

$$= \lim_{x \rightarrow 0^+} \left(\frac{\ln x}{1/x} \right) \quad \text{"}\frac{\infty}{\infty}\text{"}$$

$$= \lim_{x \rightarrow 0^+} \left(\frac{\frac{d}{dx} \ln x}{\frac{d}{dx} (1/x)} \right) = \lim_{x \rightarrow 0^+} \left(\frac{1/x}{-1/x^2} \right)$$

$$= \lim_{x \rightarrow 0^+} (-x) = 0$$

$$\ln L = 0 \Rightarrow L = e^0 = \boxed{1}$$

Example Find $\lim_{x \rightarrow 0} (1+x)^{2/x}$

$$L = \lim_{x \rightarrow 0} (1+x)^{2/x}$$

$$\ln L = \lim_{x \rightarrow 0} \left(\ln \left((1+x)^{2/x} \right) \right) = \lim_{x \rightarrow 0} \left(\frac{2}{x} \cdot \ln(1+x) \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{2 \ln(1+x)}{x} \right) \quad \text{"0/0"}$$

L'Hospital's

$$= \lim_{x \rightarrow 0} \left(\frac{2 \frac{d}{dx} (\ln(1+x))}{\frac{d}{dx} (x)} \right) = \lim_{x \rightarrow 0} \frac{2 \cdot \frac{d}{dx} (1+x)^{-1}}{1}$$

$$= \lim_{x \rightarrow 0} \frac{2}{1+x} = \boxed{2}$$

direct
subst.

$$\ln L = 2 \Rightarrow \boxed{L = e^2}$$

side note:

classic limit:

$$\boxed{\lim_{x \rightarrow 0} (1+x)^{1/x} = e}$$

$$\lim_{x \rightarrow 0} (1+x)^{2/x} = \lim_{x \rightarrow 0} \left((1+x)^{1/x} \right)^2 = e^2$$